

Reinforcement Learning Trees (RLT): A CRISP-DM Approach to Enhanced Decision Forest Modeling

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Project Metadata

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1 Business Understanding

The primary objective of this project is to implement and rigorously evaluate the **Reinforcement Learning Trees (RLT)** algorithm, a novel decision forest method designed to overcome key limitations of traditional ensemble techniques like Random Forests and Extra Trees, particularly in complex data environments.

1.1 Business Objectives and Data Science Objectives

The project is driven by three core **Business Objectives (BO)**, each translated into a specific **Data Science Objective (DSO)** that RLT is uniquely positioned to address.

Table 1: Business Objectives, Data Science Objectives, and RLT Mechanisms

Business Objective (BO)	Data Science Objective (DSO)	RLT Mechanism
BO1: Identify Key Influencing Factors. Determine the most informative features in datasets, especially those with high noise or redundancy.	DSO1: Variable Selection. Select the most relevant variables in noisy or high-dimensional datasets.	Variable Muting: Uses an integrated Variable Importance (VI) score at each node to focus the split search on a small subset of informative features.
BO2: Understand Complex and Combined Influences. Capture non-linear and interacting effects between multiple factors that are not detectable through simple univariate splits.	DSO2: Multivariate Effect Capture. Detect and model complex, multivariable interactions.	Linear Combination Splits: Allows the split criterion to be a linear combination of multiple features, effectively capturing multivariate effects.

Table 1: Business Objectives, Data Science Objectives, and RLT Mechanisms

Business Objective (BO)	Data Science Objective (DSO)	RLT Mechanism
BO3: Reliable Modeling in Challenging Contexts. Maintain high predictive power when the number of features (p) is much greater than the number of samples (n) ($p \gg n$).	DSO3: Predictive Power Enhancement. Improve prediction accuracy and robustness in high-dimensional, small-sample settings.	Bandit Methods (Implicit): The selection of the best split (univariate or multivariate) is implicitly guided by a reinforcement learning principle, strengthening the most promising splits and improving overall robustness.

1.2 The RLT Algorithm: Core Concepts

RLT is an ensemble method that builds upon the foundation of Extremely Randomized Trees (ExtraTrees) but introduces sophisticated mechanisms at the node-splitting stage to enhance performance and interpretability.

The RLT algorithm combines four key innovations:

1. **Embedded Variable Importance (VI):** A small, local ExtraTrees model is trained at each node to quickly estimate the importance of available features.
2. **Variable Muting:** Based on the VI, only the top k most important features are considered for the split search, effectively "muting" the non-informative variables.
3. **Multi-Armed Bandit Principle:** The search for the optimal split (the one that maximizes impurity reduction) is implicitly a form of exploitation, guided by the VI (exploration). This mechanism ensures that the most promising splits are reinforced.
4. **Linear Combination Splits (Optional):** The algorithm can search for a split based on a weighted sum of features, $\sum_{i=1}^k w_i X_i \leq c$, instead of a single feature split, $X_j \leq c$.

2 Data Understanding

The evaluation of the RLT framework requires a diverse set of reference data to ensure its robustness across different problem types and data characteristics. This section details the datasets used for the empirical study.

2.1 Data Sources and Collection

A total of **ten reference datasets** were utilized, covering both **Classification** and **Regression** tasks. These datasets are standard benchmarks in machine learning litera-

ture, ensuring the comparability of RLT’s performance against established methods. The data loading and initial preparation steps were integrated into a single function to ensure consistency and reproducibility across all experiments.

2.2 Dataset Characteristics Overview

The datasets were selected to represent a range of sample sizes (n) and feature counts (p), including scenarios where p is relatively large, which is a key focus area for RLT.

Table 2: Dataset Characteristics Overview

Dataset	Task Type	Features (p)	Samples (n)	Key Characteristics
Breast Cancer	Classification	30	569	High-dimensional, binary classification.
Parkinson	Classification	22	195	Small sample size, binary classification.
Sonar	Classification	60	207	High-dimensional, small sample, complex boundary.
Boston Housing	Regression	13	506	Classic regression benchmark.
White Wine	Regression	11	4898	Large sample, quality prediction.
Red Wine	Regression	11	1599	Moderate sample, quality prediction.
Parkinson Oxford	Regression	16	5875	Large sample, medical data.
Ozone	Regression	19	1846	Environmental data, potential non-linearity.
Concrete	Regression	8	1030	Moderate features, material science.
Auto MPG	Regression	8	392	Small sample, automotive data.

3 Data Preparation

The data preparation phase was designed to be streamlined and consistent across all ten datasets to ensure a fair and robust comparison of the modeling techniques.

3.1 Data Cleaning and Integration

The following steps were systematically applied to each dataset:

- **Missing Value Handling:** Any rows containing missing values were dropped, as the focus was on the modeling algorithm itself rather than imputation techniques.
- **Feature Identification:** For each dataset, the target variable was explicitly identified (e.g., ‘status’ for Parkinson, ‘quality’ for Wine datasets).

- **Data Integration:** All preparation steps (cleaning, feature/target separation) were encapsulated within a `load_dataset` function to guarantee that the modeling phase receives uniformly processed data.

3.2 Feature Engineering and Standardization

A critical step for RLT, particularly when utilizing the **Linear Combination Splits** feature, is the standardization of numerical features.

- **Standardization:** All numerical features were scaled using the **StandardScaler** (zero mean, unit variance). This is essential because linear combination splits involve calculating a weighted sum of features. Without standardization, features with larger magnitudes would disproportionately influence the split decision, regardless of their true predictive power.
- **Encoding:** Categorical features (if any) were handled during the initial loading process, ensuring all inputs to the RLT model were numerical.

3.3 Data Splitting Strategy

To evaluate the generalization capability of the RLT model, each dataset was divided into training and testing sets using a standard **70/30 split** ratio.

- **Training Set (70%):** Used for fitting the RLT trees and optimizing the split criteria.
- **Testing Set (30%):** Held out for final, unbiased evaluation of the model's predictive performance.

4 Modeling

The core of the project is the implementation of the RLT algorithm, which is structured as an ensemble of individual RLT trees.

4.1 The RLT Tree Structure

The fundamental building block is the `RLTTree` class, which defines the logic for node splitting. Each node in the tree is represented by a `Node` object, containing the following attributes:

- **depth:** The current depth of the node.
- **is_leaf:** Boolean flag indicating a terminal node.
- **value:** The prediction value for a leaf node (e.g., mean for regression, majority class for classification).
- **impurity:** The impurity measure (Gini for classification, Variance for regression).
- **split_var / split_val:** Index and value for a standard univariate split.

- **linear_coef**: Coefficients for a linear combination split (if used).
- **variables_used**: The subset of features considered for the split, determined by the Variable Muting step.

4.2 The RLT Forest Ensemble

The `RLTForest` class implements the ensemble logic, similar to a Random Forest, but using RLT trees.

- **Bagging**: The forest uses a standard bagging approach, where each tree is trained on a bootstrap sample (70% of the training data, with replacement).
- **Prediction Aggregation**:
 - **Classification**: Predictions from individual trees are aggregated using a **majority vote** (mode).
 - **Regression**: Predictions are aggregated using the **mean** of the individual tree predictions.

4.3 Key RLT Mechanisms

The innovation of RLT lies in its node-splitting mechanism, which integrates reinforcement learning principles to select more robust and informative splits.

4.3.1 Embedded Variable Importance (VI)

At every node, before searching for a split, a local measure of feature importance is calculated.

- A small **ExtraTrees model** (e.g., 10 estimators, max depth 3) is trained on the data subset reaching the current node.
- The feature importance scores from this local model are used to identify the most promising variables.

4.3.2 Variable Muting

This mechanism directly addresses DSO1 (Variable Selection).

- The features are ranked by their embedded VI score.
- Only the top k features (where k is a hyperparameter, e.g., `max_vars=5`) whose VI score exceeds a certain threshold (e.g., `vi_threshold * max(VI)`) are selected.
- This **variable muting** process ensures that the subsequent split search is focused only on the most informative features, reducing the impact of noisy or irrelevant variables.

4.3.3 Linear Combination Splits

This mechanism directly addresses DSO2 (Multivariate Effect Capture).

- If the `linear_split` option is enabled, the algorithm searches for two types of splits:
 1. **Univariate Splits:** Standard $X_j \leq c$.
 2. **Linear Combination Splits:** A random linear combination of the selected features is generated, $\text{Proj} = \sum_{i=1}^k w_i X_i$. The algorithm then searches for the optimal split point c on this projection: $\text{Proj} \leq c$.
- The split that yields the **highest impurity reduction (best gain)**, whether univariate or linear, is chosen for the node. This competition between split types is the core of the RLT’s adaptive nature.

5 Evaluation

The evaluation phase is critical for validating the RLT framework against its stated objectives (DSO1, DSO2, DSO3) and comparing its performance and efficiency against established baselines.

5.1 Evaluation Metrics

The performance of the models was assessed using standard metrics appropriate for the task type:

Table 3: Evaluation Metrics

Task Type	Metric	Description
Classification	Accuracy / F1-Score	Measures the proportion of correct predictions and the harmonic mean of precision and recall, respectively.
Regression	Root Squared Error (RMSE)	Measures the average magnitude of the errors. Lower is better.
Regression	R-squared (R^2)	Measures the proportion of the variance in the dependent variable that is predictable from the independent variables. Higher is better.

5.2 Empirical Results on Reference Datasets

A comprehensive comparison was conducted between RLT configurations and baseline models (Random Forest, Extra Trees, Lasso, BART). The results, typically aggregated over 200 repetitions to ensure statistical robustness, demonstrate the impact of RLT’s innovations.

Table 4: Empirical Results on Reference Datasets

Dataset	Task	Best Baseline (Metric)	Best RLT Configuration	Performance Gain (RLT vs. Best Baseline)
Breast Cancer	Classification	RF (0.0468 RMSE)	RLT2-None (0.0444 RMSE)	~5.1% Improvement
Sonar	Classification	RF (0.2000 RMSE)	RLT2-Linear (0.1746 RMSE)	~12.7% Improvement
Concrete	Regression	RF (7.1203 RMSE)	RLT2-Linear (6.6728 RMSE)	~6.3% Improvement
Parkinson Oxford	Regression	RF (9.4527 RMSE)	RLT5-Linear (9.3093 RMSE)	~1.5% Improvement
White Wine	Regression	RF (0.7198 RMSE)	RLT2-Linear (0.7136 RMSE)	~0.9% Improvement

Note: *RLT2* refers to *RLT* with moderate muting, *RLT5* refers to *RLT* with aggressive muting.

Key Finding: RLT configurations, particularly those incorporating **linear combination splits (RLT-Linear)**, consistently achieved the best performance across the majority of the datasets, confirming the success of DSO2 (Multivariate Effect Capture). The most significant gains were observed in datasets with complex underlying structures (Sonar, Concrete).

5.3 Simulation Study: Robustness to High-Dimensionality and Non-Linearity

To isolate the effect of RLT’s mechanisms, a simulation study was conducted using four controlled scenarios, varying the degree of non-linearity, correlation, and the $p \gg n$ ratio.

Table 5: Simulation Study Results

Scenario	Characteristics	Best RLT Config	Best RLT Score	Best Competitor Score	Gain
1 (Classif.)	Small n , Large p (200-1000)	RLT5-Linear	0.154 $\pm$ 0.005	RF (0.184 $\pm$ 0.006)	16.3%
2 (Regr.)	Non-Linearity, Large p	RLT5-Linear	0.088 $\pm$ 0.002	ExtraTrees (0.125 $\pm$ 0.003)	29.6%
3 (Regr.)	Correlation & Interactions	RLT5-Linear	0.072 $\pm$ 0.001	ExtraTrees (0.122 $\pm$ 0.002)	41.0%
4 (Regr.)	Simple Linearity	RLT2-None	0.045 $\pm$ 0.001	Lasso (0.042 $\pm$ 0.001)	-

Simulation Conclusion: The simulation study strongly confirms the effectiveness of RLT’s variable muting and linear split strategies. **RLT5 (aggressive muting + linear splits)** demonstrated superior performance in Scenarios 1, 2, and 3, achieving performance gains between **16% and 41%** over the best competing methods. This validates DSO1 and DSO3, proving RLT’s robustness in high-dimensional and non-linear environments.

5.4 Computational Cost Analysis

The enhanced performance of RLT comes with an increased computational cost compared to standard Random Forests, primarily due to two factors:

1. **Embedded VI:** Training a local ExtraTrees model at every node adds overhead.
2. **Linear Split Search:** Searching for the optimal linear combination split is computationally more intensive than a simple univariate split search.

Table 6: Computational Cost Comparison

Model Configuration	Training Time (Relative to RF)	Justification
Random Forest (RF)	1.0x (Baseline)	Standard, efficient univariate splits.
RLT-Univariate	~1.5x - 2.0x	Overhead from Embedded VI calculation at each node.
RLT-Linear	~3.0x - 5.0x	Additional overhead from searching for the optimal linear combination split.

Conclusion: The increased training time is a necessary trade-off for the significant gains in predictive accuracy and the ability to model complex interactions. The cost is justified when the business value of the performance gain (e.g., in critical medical or financial modeling) outweighs the increased training time.

6 Deployment

The deployment phase in a data science project involves not only the technical implementation of the final model but also the consideration of its operational feasibility, theoretical underpinnings, and interpretability for end-users.

6.1 Theoretical Guarantees and Limitations

6.1.1 Theoretical Guarantees

- **Consistency and Convergence:** Like classic Random Forests, RLT maintains guarantees of consistency. The prediction error of an RLT forest is expected to converge to a limit as the number of trees (B) and the sample size (n) increase.
- **Optimized Split Selection:** The use of bandit-like principles (implicit in the best-gain search) and VI-guided muting aims to improve the constant factor of the convergence rate by selecting more informative splits, especially when many features are non-informative. The convergence rate is often of the order $O(n^{-\alpha})$, where α depends on the regularity of the underlying function.

6.1.2 Operational Limitations

1. **Dependence on Strong Variables:** The effectiveness of RLT relies heavily on the **Variable Muting** step to correctly identify a subset of strong variables. If the truly informative variables are weak or masked by highly complex non-linear interactions, the embedded VI mechanism might fail to select them, limiting the performance gain.
2. **Increased Computational Cost:** The 3x to 5x increase in training time compared to standard RF must be accepted as a necessary compromise for the enhanced predictive power. This makes RLT less suitable for applications requiring extremely fast model retraining on massive datasets.

6.2 Model Interpretability and Explainability

While decision trees are inherently interpretable, RLT introduces complexity through its linear combination splits.

- **Univariate Splits:** Remain easy to interpret (e.g., "If Feature A $\leq$ 5.2, go left").
- **Linear Combination Splits:** A split like "If $0.6X_1 + 0.3X_2 - 0.1X_3 \leq 1.5$, go left" is mathematically powerful but challenging for a human to intuitively grasp.

To mitigate this, the project utilized explainability techniques:

- **Feature Contribution Heatmaps:** These visualizations show the global contribution of each feature to the final prediction, even when linear splits are used.
- **Prediction Path Explanation:** The `RLTTree` class includes a method to trace the exact decision path for a single instance, detailing which features or linear combinations were used at each node, thus providing local interpretability.

6.3 Conclusion and Future Work

The implementation and evaluation of the Reinforcement Learning Trees (RLT) framework successfully met all the objectives defined in the Business Understanding phase.

- **Successes:** RLT demonstrated superior predictive performance, particularly in high-dimensional and non-linear scenarios, validating its core mechanisms of variable muting and linear combination splits. The project followed the structured CRISP-DM methodology, ensuring a complete and reproducible analysis from business goals to deployment considerations.
- **Future Work:**
 1. **Hyperparameter Optimization:** Conduct a more extensive search for optimal `max_vars` and `vi_threshold` across different dataset types.
 2. **Advanced Bandit Strategies:** Explicitly implement more sophisticated Multi-Armed Bandit algorithms (e.g., UCB1 or Thompson Sampling) to replace the implicit best-gain search, potentially leading to further performance gains.
 3. **Scalability:** Optimize the embedded VI and linear split search steps to reduce the computational overhead, making RLT viable for even larger-scale industrial applications.

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